

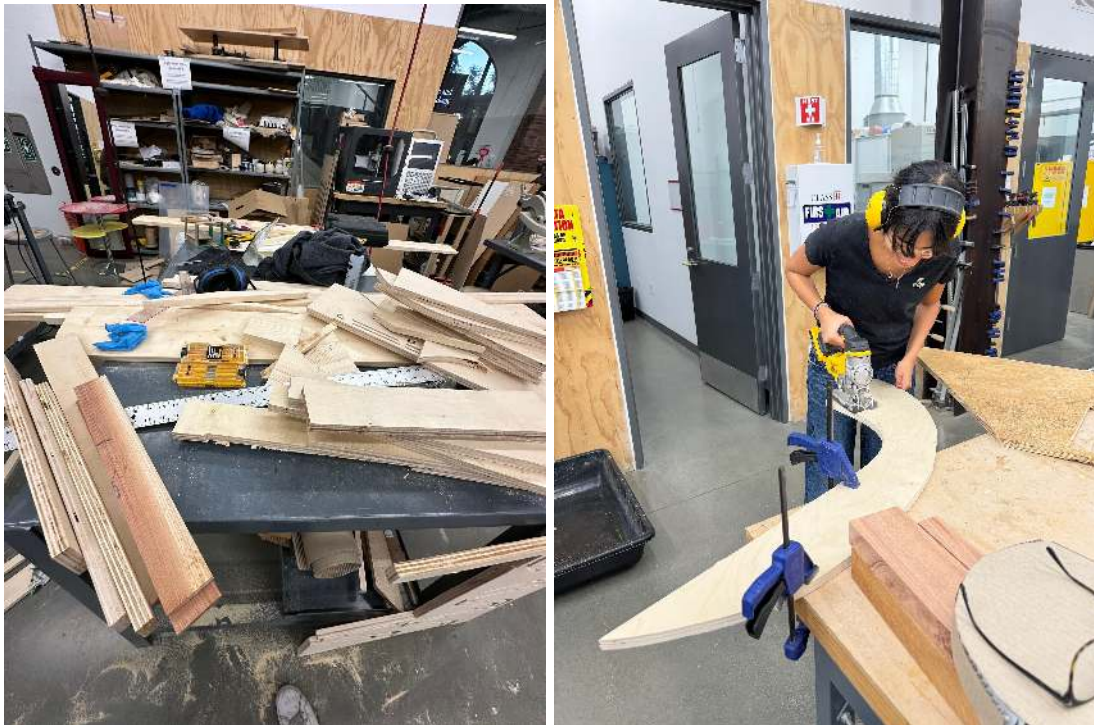
Final Project: Floor Rocking Chair for Elementary School Classrooms



For my final project I wanted to make a piece of furniture that students in the elementary classroom (Alexander Science Center) I taught Python in this semester could use. Every week I would go in the classroom and notice the fun furniture the students would sit in during their free time. I grew so close to these kids this semester and wanted to make a parting gift for them that would serve as a memory of the time we spent together.

For this project, I wanted to use 100% scrap wood to promote sustainability so that it is inspiring for the students as well, since their curriculum is already very centered around seeing USC students as role models in their career. It's also always nice to keep a budget for myself of course.

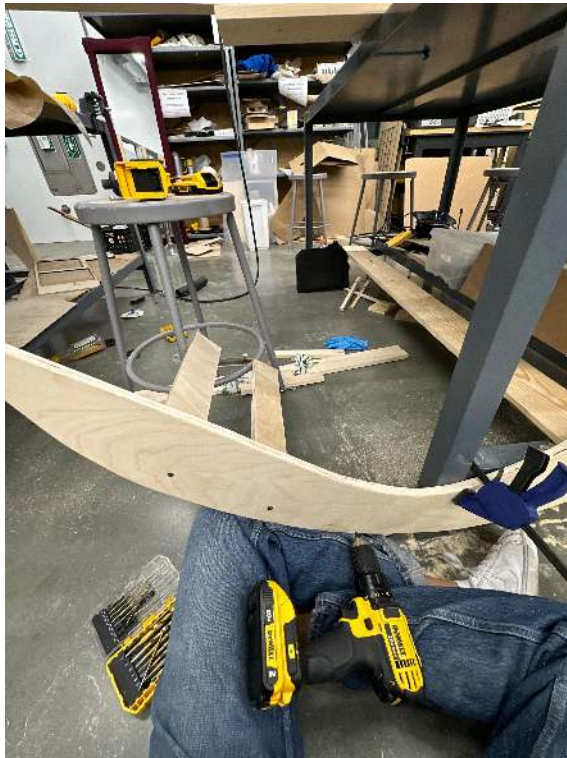
I found plywood in the creator studio and found two large pieces of cut up wood that perfectly fit two side planks diagonally across. I drew one out, cut it with a jigsaw, and traced that onto another scrap wood to cut another symmetrical frame.



For the planks, it was very difficult to find enough wood of the same length and width. Either I would cut them all to the same width and have shorter pieces, or I would have a smaller amount of longer pieces of wood to support the chair. What I decided to do was keep different widths, but all a good medium length so that a small kid could comfortably fit in it.

I had many iterations of the idea throughout the making process. I wanted to drill into the wood because it seemed like the sturdiest option. However some wood was weaker than others so when I drilled some wood snapped apart on the inside. It still held together, but it was definitely a hazard. I wanted to use joinery, but the only one I could think of was using dowels, but the maker space dowel kit was lost and due to time restraints I could not feasibly start over to complete that task, especially since I did not have

more wood to restart. However in the future, I think if I did drill, it would be more sturdy to use stronger wood like maple wood and to put more screws, two on one and pocket screws on the other side.



Due to the wood snapping slightly, I wondered if I could turn the rocking chair into a beach chair by adding two planks supporting the back. I found a series of scrap wood to make a rectangle slit for the planks of wood to slide into, a type of joinery I wanted to try for a sturdier structure.



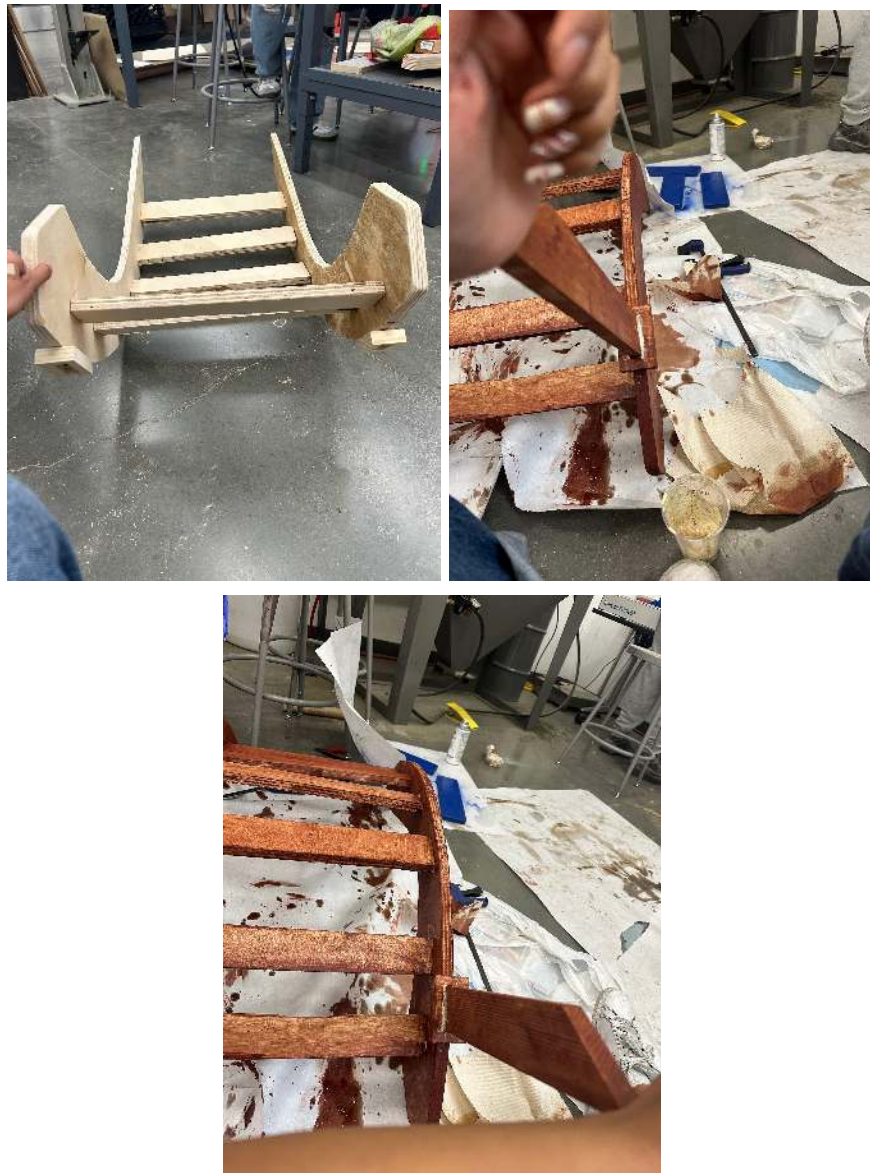
This took three attempts. The first one I tried drilling a bunch of holes with a drill bit and using the same drill to cut out the wood in the slit. This terribly ruined the wood.

I started over by using a drill bit but using a jigsaw instead to cut a slit through all the holes. This was better, but still uneven and did not hold the planks well.



My third attempt was the best, but still not enough. I used the pull down dole drill in the wood room to make three big but very close together holes, and used a dremel to shave away the extra wood edges into a rectangle. However the bottom was uneven no matter how much sanding I did, and the slit was still not snug enough for the wooden planks to hold.

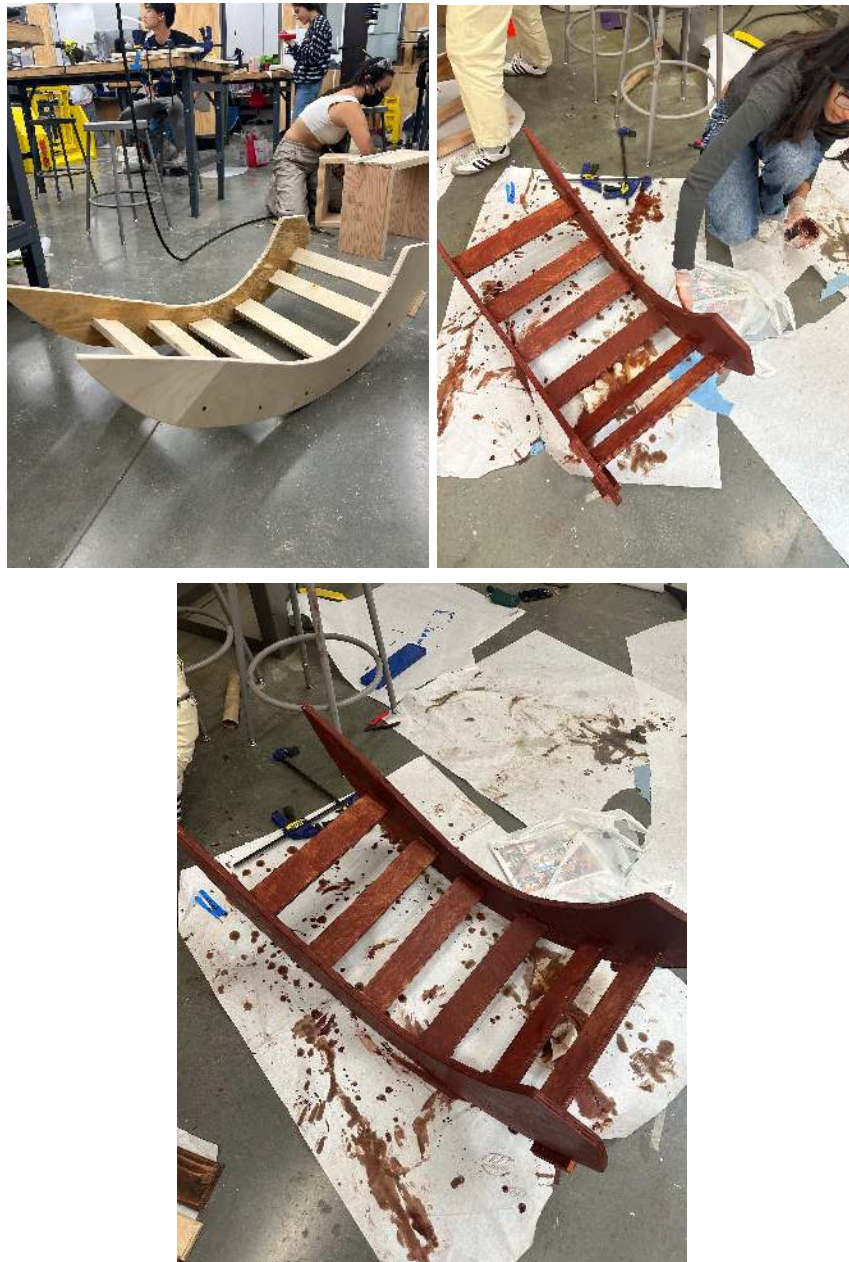
I decided to try gluing the wood into the slits I made after the hours of effort I already put into it, but the wood did not stick and was not a smart choice structurally either since it would be taking on a lot of weight if someone sat on it. I even tried adding more glue mixed with sawdust but that did not work either, and it was difficult to let it dry because resting it was not enough pressure, and there were no supports that could clasp onto the structure I had. I sat there holding it for an hour, very determined to make it work, but it never dried so I ended up getting rid of this extra structure.



I found that the most difficult thing about wood for me is making sure to get precise measurements. The slit for the planks was not deep enough to have enough structure anyway, so I decided to revert back to my original idea and keep the floor rocking chair.

Although I had a lot of unsuccessful iterations, I learned a lot about the process of building functional furniture, learning to think about safety, structure, symmetry and material used.

Finally, I finished the wood with a dark wood stain so that it gave a safe and warm vibe for the classroom, and not like an arts project (even though it felt like one).



I also added a small mattress on top to complete the look!



This project and class has inspired me to design and create products that people can use immediately, and in the future I want my own furniture and interior decorations to be handmade, now that I have learned through making myself, and by watching other people's processes in their own building projects. In the future I want to try laser cutting, using different joineries like rabbet joinery, L shaped, and dowels.

Also, if the structure did not work for the chair, I could use it as a drying rack which I thought was nice to have as a dual functionality!



Materials used:

- Plywood
- Pine wood
- Other unknown scrap wood
- Wood stain finish

Tools used:

- Table saw
- Circular saw
- Orbital sander
- Pull down hole cutter (forgot name but it's in the wood room)
- Hand paper sander
- Dremel
- Jigsaw
- Screwdriver
- Drill bit
- Impact driver
- Wood glue
- Wood stain finishing
- Supports
- Metal support